



9/17/25

Fermentation Processes for Production of Antibiotics (Penicillin)

Kendall Johnson, Joshua Olusoji,
Jazmin Aca, Qussay Suleiman



UNIVERSITY OF
ILLINOIS CHICAGO



Table of Contents

1. Introduction
2. Process Description
3. Design Basis
4. Flow Diagrams
5. Bibliography

An aerial photograph of a university campus, likely the University of Illinois at Chicago (UIC), showing a large, modern building complex with multiple wings and courtyards. The image is overlaid with a semi-transparent blue filter. Numerous people are visible walking along the paths and courtyards. Several UIC logos are visible on the buildings and signage. The word "Introduction" is centered over the image in a large, white, sans-serif font.

Introduction

Why should we care about Penicillin?

Diseases fought with Penicillin:

- Scarlet Fever
- Meningitis
- Any gram – positive bacteria



Introduction

Deep-tank fermentation allows for the production of penicillin in a controlled, pure, and stable way.

Advancements within the production of penicillin:

- Combatting antibiotic resistant bacteria
- Improved purity of yields with new technology
- Environmental considerations

Market Analysis

There is a potential for a an approx. \$25 Billion USD market size increase within the antibiotics industry.

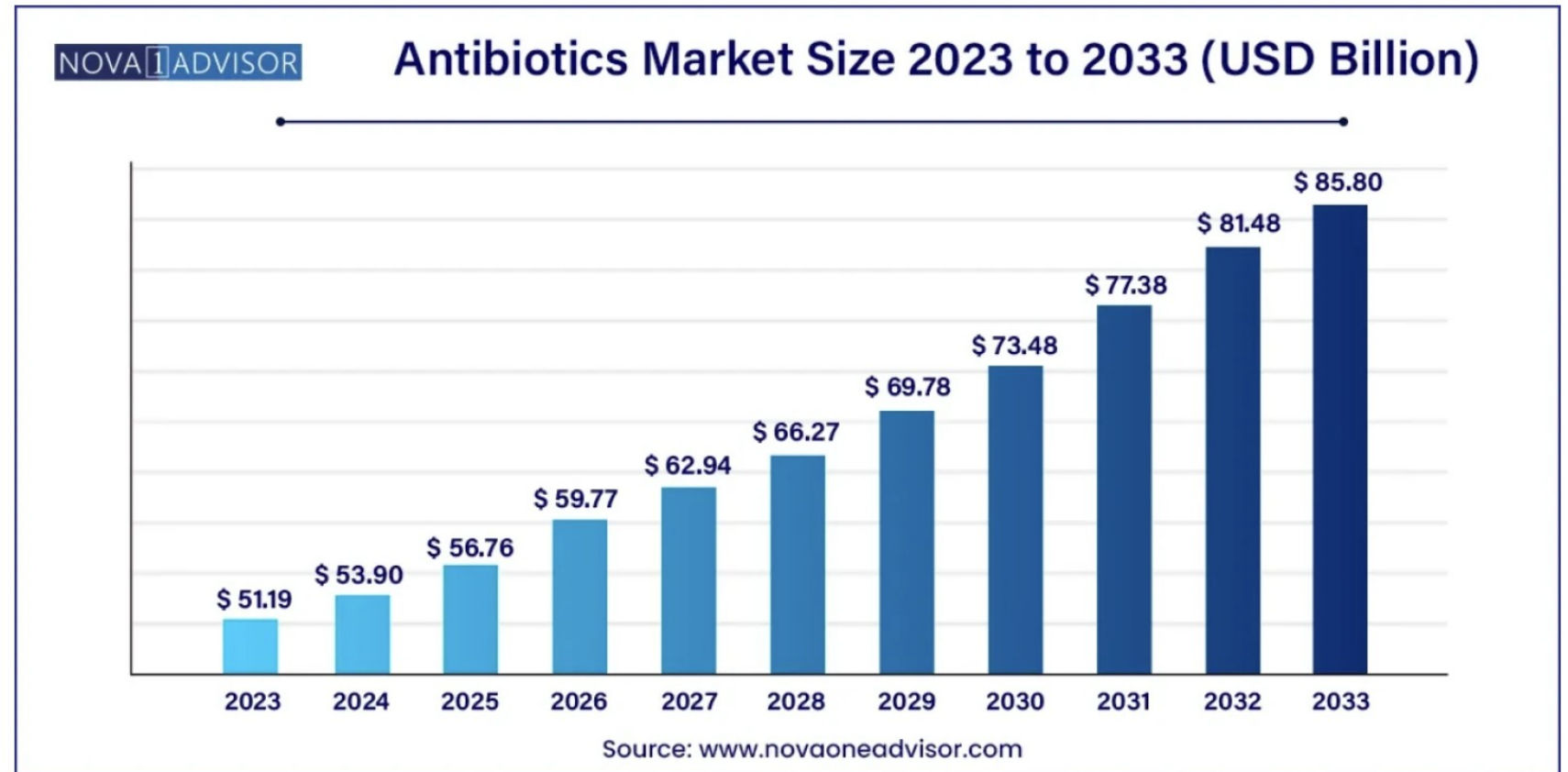


Figure from Nova 1 Advisor

An aerial photograph of a university campus, likely the University of Illinois at Chicago (UIC), is shown with a semi-transparent blue overlay. The image captures a wide pedestrian walkway lined with trees, with many people walking. In the background, a large building is visible. Several UIC logos are scattered throughout the scene, including one on a signpost and another on a building. The title 'Process Description' is centered in white text over the middle of the image.

Process Description

How is Penicillin Made?

There are many steps to producing Penicillin, but they can be summarized as follows:

- **Medium prep:** Mix nutrients, sterilize.
- **Inoculum:** Build clean seed culture.
- **Fermentation:** Aerated tank; mold grows, then makes penicillin.
- **Recovery:** Remove cells, extract, purify to crystals.

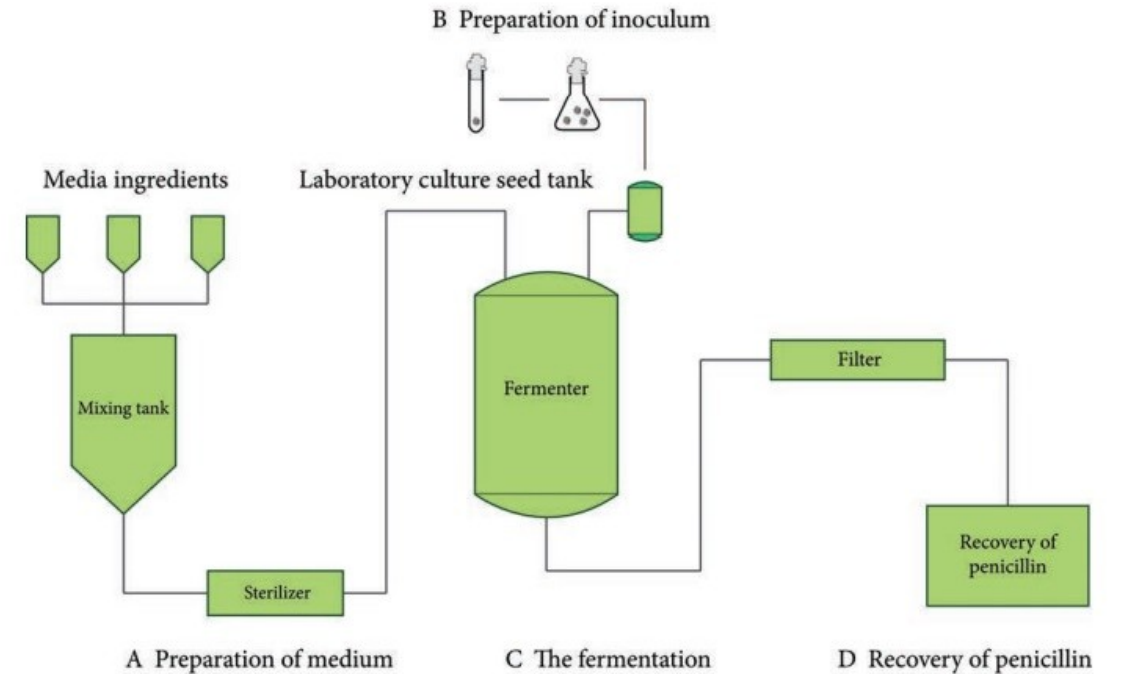


Figure 1: Production of Penicillin

Chosen process

- Penicillin produced from combined upstream & downstream processes
- No real competing upstream processes to produce penicillin, fermentation is essential
- Different techniques to separate biomaterial from crude penicillin
- Waste treatment varies (implementation of optimization in recycle streams)
- Improper waste treatment can have too much penicillin leaving facility & can lead to antibiotic resistance

Upstream Process	Downstream Process
Aerobic Conditions	Fungal biomass, nutrients, crude penicillin separated
pH= 6.0 – 6.5	Centrifugal Separator
Temp = 25 °C	Chromatography
Precursors Added (phenylacetic acid etc)	Precipitation
	Crystallization

An aerial photograph of a university campus, likely the University of Illinois at Chicago (UIC), showing a large, open plaza with many people walking. The image is overlaid with a semi-transparent blue filter. The text "Design Basis" is centered in the middle of the image in a white, sans-serif font.

Design Basis

Plant information

Plant size (industrial fed-batch)

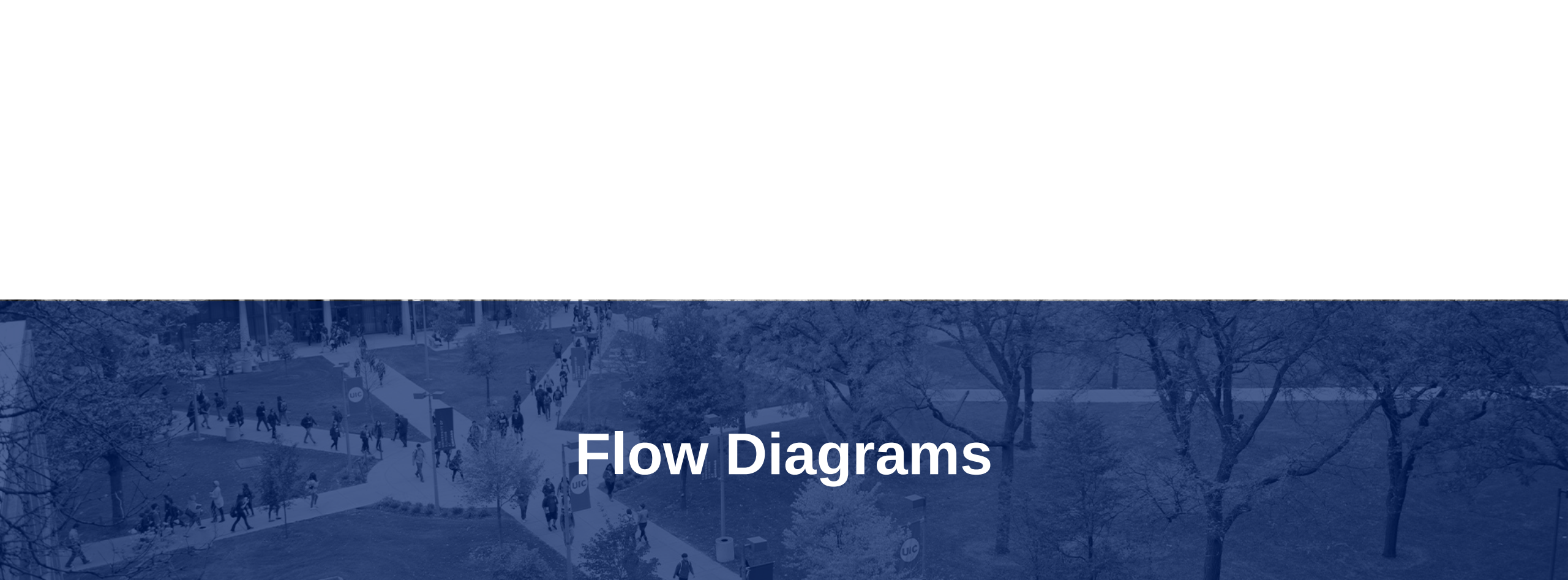
- Fermenters: 4 × 120 m³ stirred-tank reactors (plus seed tanks).
- Throughput: titer ~40 g/L, overall recovery ~90% → ~4.3 t Pen-G per batch per tank.
- Annual output (200 total batches/year across all 4 tanks): ~860 t Penicillin G per year.

Plant location

- Location: India
- Why here (quick hits):
 - Cost-competitive labor and utilities.
 - PLI incentives supporting domestic API manufacturing.
 - Abundant inputs from large sugar and dairy sectors.
 - Strong port and logistics for export.

Raw Materials

Material	Per batch	Per year
Lactose (carbon source)	3,600 kg	720,000 kg
Corn steep liquor (CSL) solids	3,600 kg	720,000 kg
Calcium carbonate (CaCO ₃)	120 kg	24,000 kg
Phenylacetic acid (PAA)	60 kg	12,000 kg
Process water (medium volume)	120,000 L	24,000,000 L



Flow Diagrams

Block Flow Diagram

Inoculation

- Lots of nutrients are required to create an environment where Penicillium grows!

Fermentation Reaction

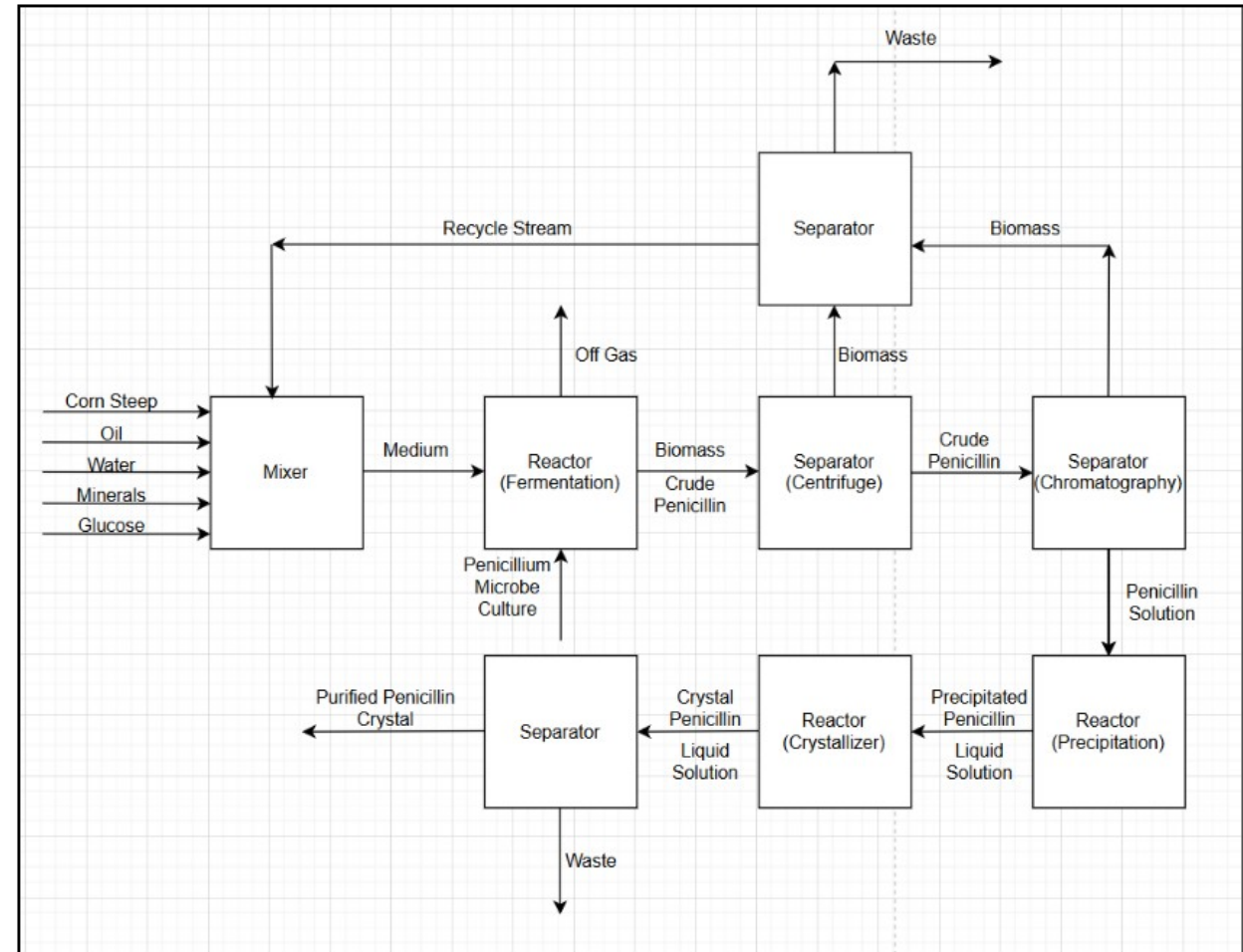
- Fermentation takes multiple days/weeks, requiring a batch process & reactor
- Biomass (mycelium metabolites) are formed in addition to the penicillin, which are removed via rotary filtration/centrifugation

Purity & Further Separation

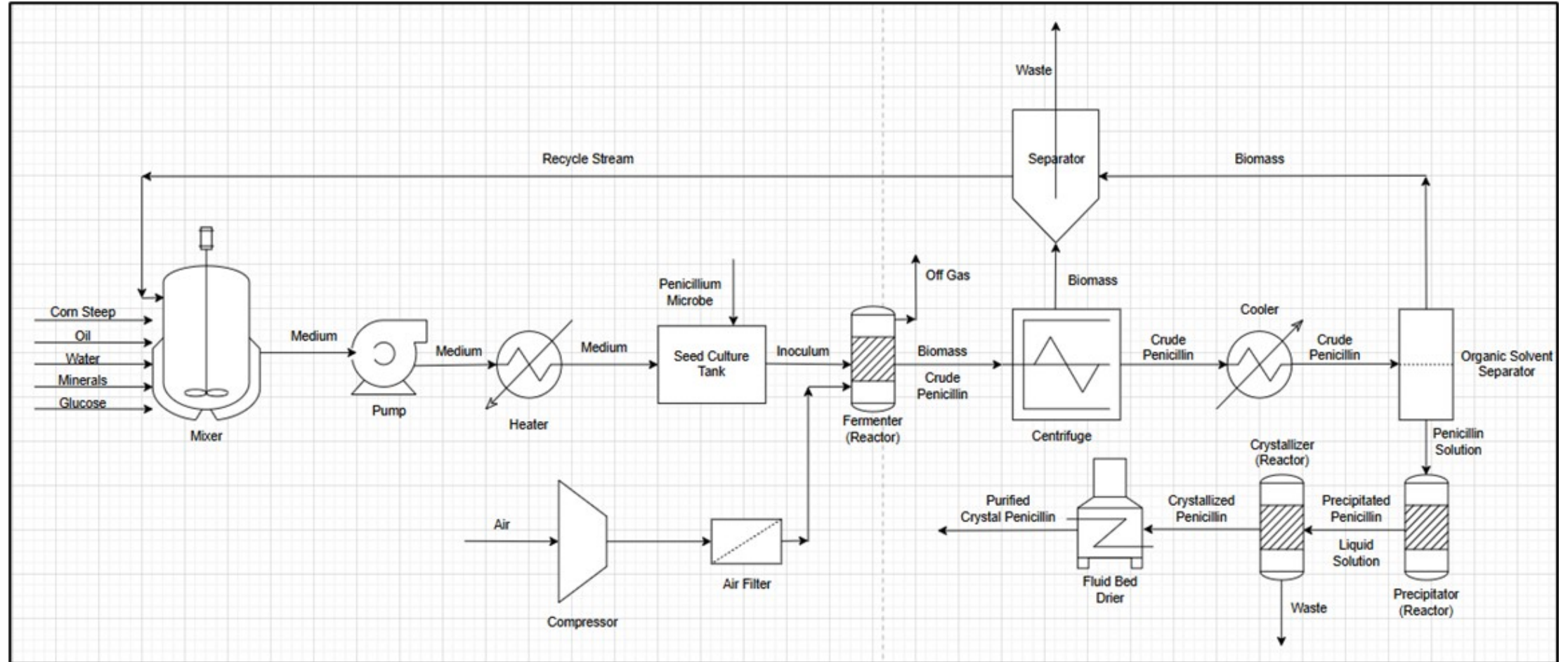
- Chromatography, precipitation reactions, and recrystallization are utilized to increase the purity of the penicillin formed

Recycle Streams

- In more complex processes, water & other nutrients are recycled for sustainability reasons!

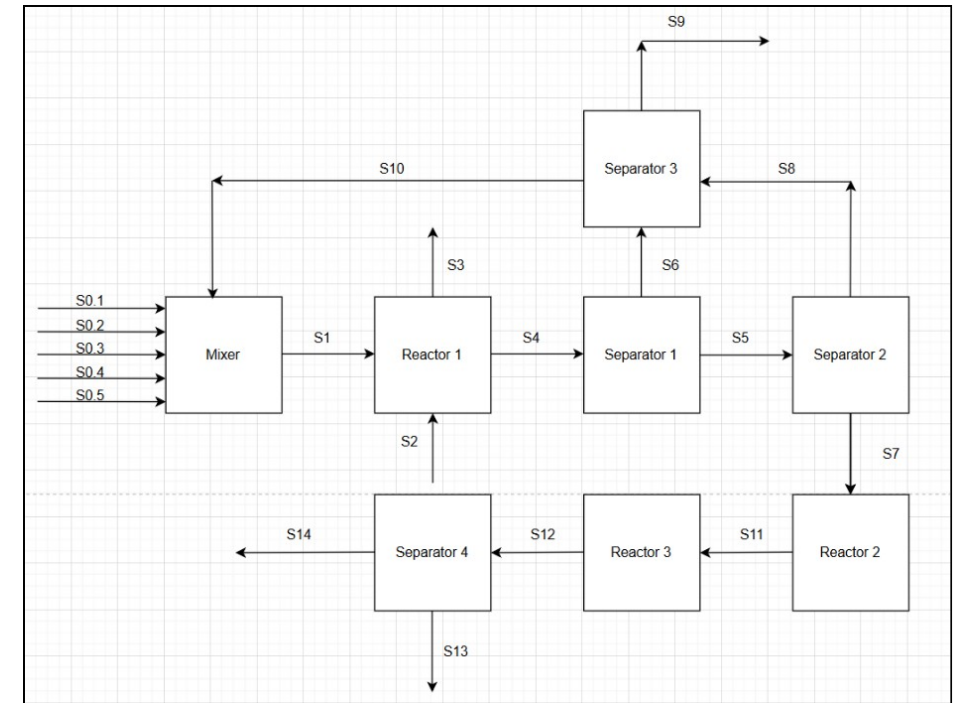


Process Flow Diagram



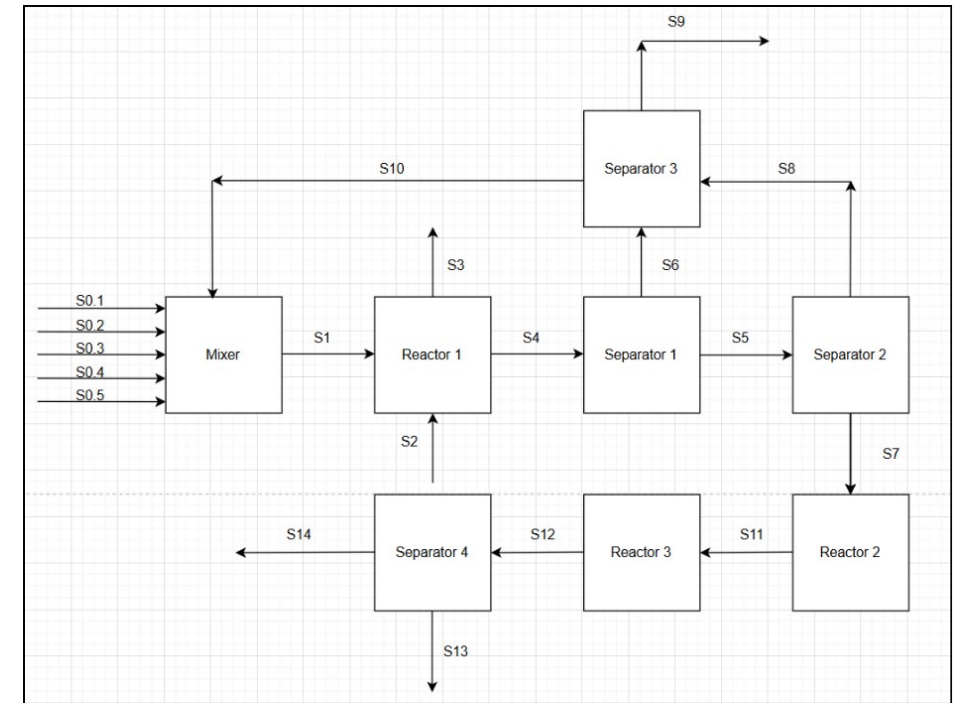
Preliminary Material Balance

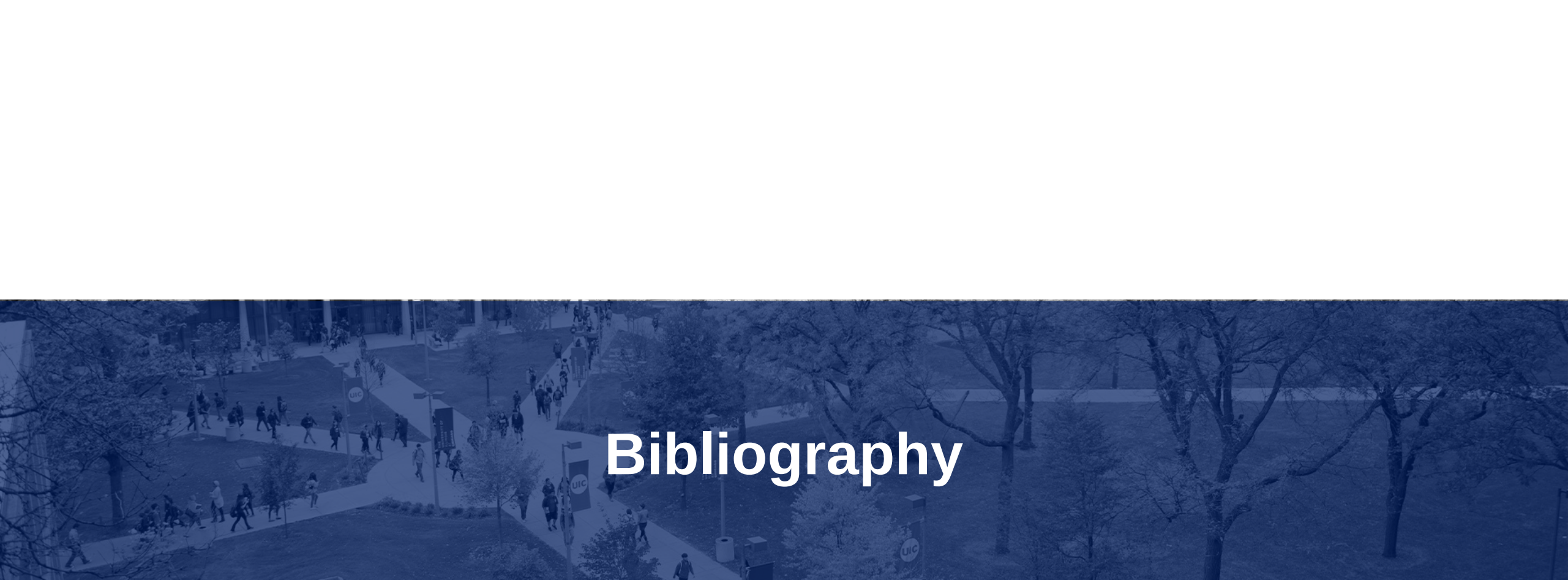
Unit Operation	Material Balance	Explanation
Mixer	$S_0 + S_{10} = S_1$ (Nutrients) + (Recycle Stream) = (Penicillium Medium)	Corn steep, oil, water, minerals, and glucose are added to the system to create a nourishment rich environment for Penicillium to grow. Additionally, some water is recycled for sustainability reasons.
Reactor 1	$S_1 + S_2 = S_3 + S_4$ (Pen. Medium) + (Pen. Microbe) = (Off Gas) + (Crude Penicillin & Biomass)	The medium is heated and then implanted with penicillium, creating an inoculum. Fermentation begins to create crude penicillin and excess biomass. Additionally, CO ₂ is formed and released as off-gas.
Separator 1	$S_4 = S_5 + S_6$ (Crude Pen. & Biomass) = (Crude Pen.) + (Biomass)	A centrifuge is used to separate crude penicillin from excess biomass formed during the fermentation.
Separator 2	$S_5 = S_7 + S_8$ (Crude Pen.) = (Penicillin Solution) + (Biomass)	The crude penicillin is cooled and placed in organic solvent chromatography, further separating excess biomass from the penicillin.



Preliminary Material Balance

Unit Operation	Material Balance	Explanation
Separator 3	$S6 + S8 = S9 + S10$ (Biomass) + (Biomass) = (Waste) + (Water)	The biomass is stripped of water and nutrients to be recycled and the remainder of it is sent out of the system as waste.
Reactor 2	$S7 = S11$ (Pen. Solution) = (Precipitated Penicillin & Liquid Solution)	Penicillin is reextracted from solution using a precipitant, creating a much purer penicillin salt.
Reactor 3	$S11 = S12$ (Precipitated Pen. & Liq. Sol.) = (Crystal Penicillin & Liq. Sol.)	A recrystallization reactor creates high-purity crystal penicillin.
Separator 4	$S12 = S13 + S14$ (Crystal Pen. & Liq. Sol.) = (Liquid Waste) + (Crystal Pen.)	Finally, a separator is used to harvest pure crystal penicillin.



An aerial photograph of a university campus, likely the University of Illinois at Chicago (UIC), showing a large, modern building complex with multiple wings and courtyards. The image is overlaid with a semi-transparent blue filter. The word "Bibliography" is centered in white text over the middle of the image.

Bibliography

Works Cited

American Chemical Society National Historic Chemical Landmarks. (2008). Penicillin Production through Deep-tank Fermentation. American Chemical Society National Historic Chemical Landmarks. <http://www.acs.org/content/acs/en/education/whatischemistry/landmarks/penicillin.html>

Department of Pharmaceuticals. (2025). Production Linked Incentive (PLI) Scheme for Promotion of Domestic Manufacturing of critical Key Starting Materials (KSMs)/ Drug Intermediates and Active Pharmaceutical Ingredients (APIs) in the Country. Department of Pharmaceuticals. https://pharma-dept.gov.in/schemes/production-linked-incentive-plischeme-promotion-domestic-manufacturing-critical-key?utm_source=chatgpt.com

Editorial Team. (2024, May 19) Penicillin production. Biology Notes Online. <https://biologynotesonline.com/penicillin-production/>

Notes for Biology. "Penicillin Production Commercially." *Notes for Biology*. Accessed September 16, 2025. <https://notesforbiology.com/penicillin-production-commercially/>

Haque, Md & Deb Nath, Nirmalendu & Johnston, Tony & Haruna, Samuel & Ahn, Jaehyun & Ovissipour, Reza & Ku, Seockmo. (2024). Harnessing biotechnology for penicillin production: Opportunities and environmental considerations. *Science of The Total Environment*. 946. 174236. 10.1016/j.scitotenv.2024.174236.

Meštrović, Dr. T. (2023, June 17) Penicillin production. News Medical Life Sciences. <https://www.news-medical.net/health/Penicillin-Production.aspx>

San, K., Stephanopoulos, G. (1989). Optimization of fed-batch penicillin fermentation: A case of singular optimal control with state constraints. *Biotechnology and Bioengineering*. <https://doi.org/10.1002/bit.260340110>

News-Medical Life Sciences. "Penicillin Production." *News-Medical*, accessed September 2025. <https://www.news-medical.net/health/Penicillin-Production.aspx>

Thank you!

Any Questions?